LECTURE 16
Finite Differences in 2D: diffusion and advection

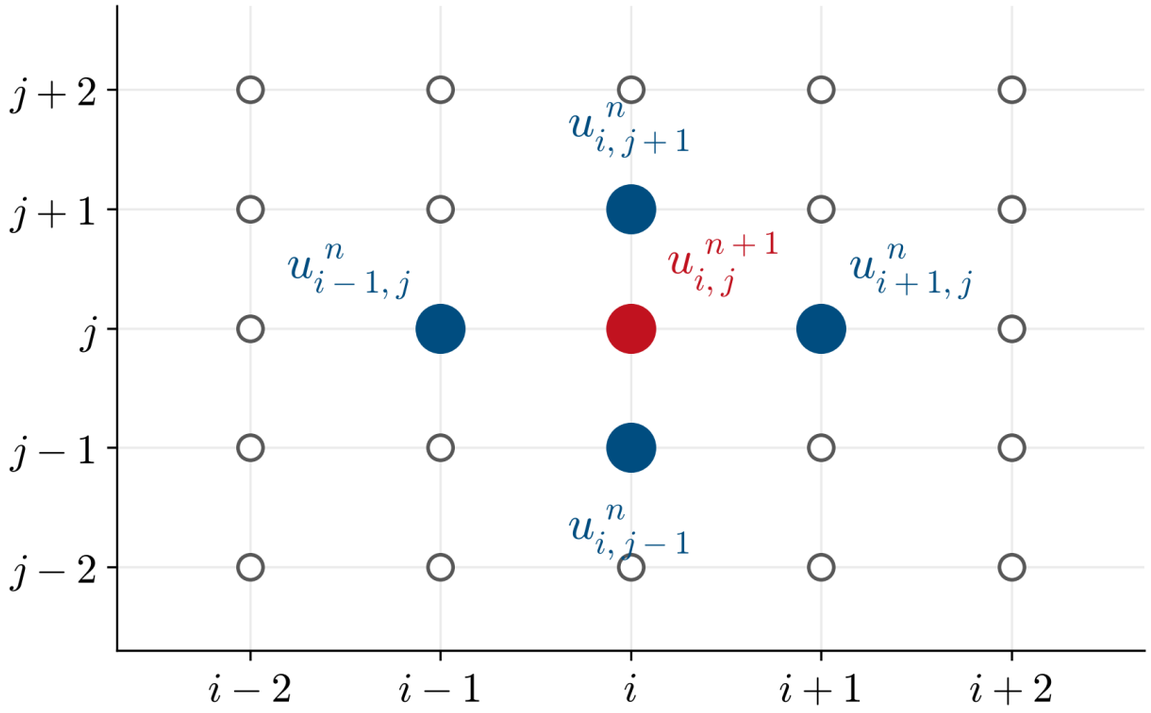
Reference: J. C. Tannehill, D. A. Anderson & R. H. Pletcher, Computational Fluid Mechanics and Heat Transfer, 2nd ed., Taylor & Francis, 1997. Sections 4.2.9 to 4.2.11.
L. A. Barba & G. F. Forsyth, 12 steps to Navier-Stokes, steps 5-7.
Wladimir E. Banda-Barragán

Finite differences in 2D

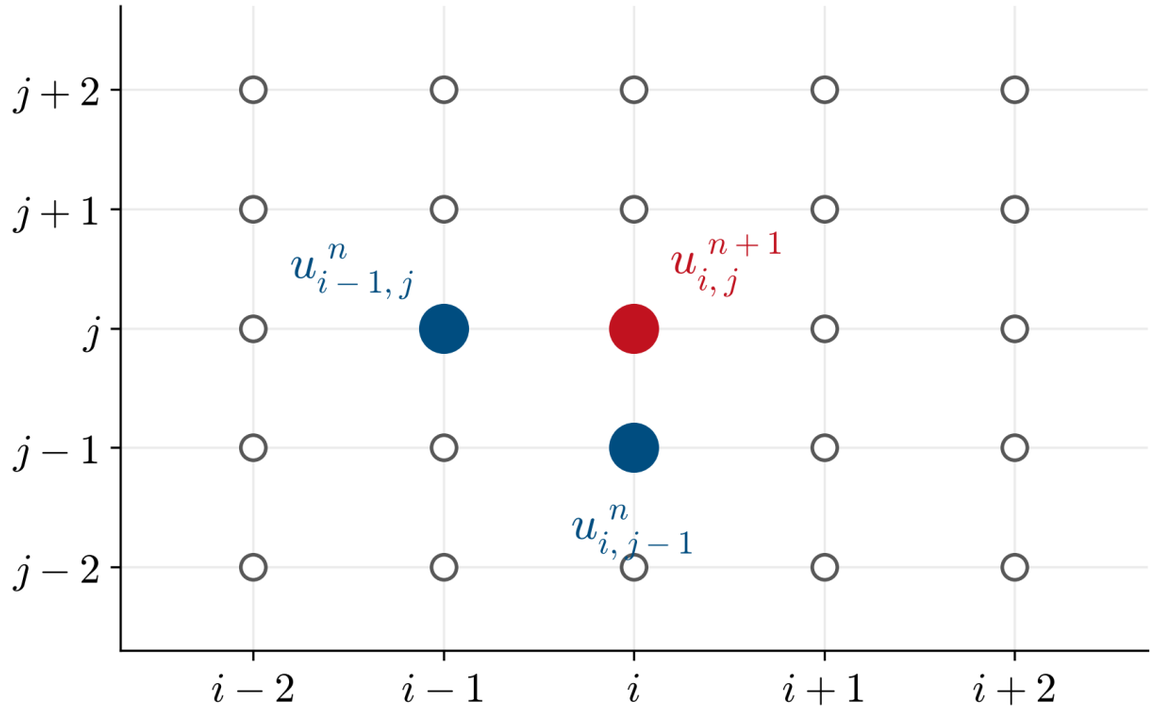
We use the finite difference formulae from 1D and apply them in each direction.

We write $u[i, j, n]$, with i for x, j for y and n for the time level.

FTCS Diffusion: 5-point stencil



Upwind advection for $c_x > 0$ and $c_y > 0$



The new value sits over the centre in red.
The values used by the scheme are in blue.

The 2D Diffusion Equation

We start from the heat equation, now with two space derivatives on the RHS:

$$\frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

We difference it forward in time and centred in space, which gives the 5-point FTCS scheme:

$$u_{i,j}^{n+1} = u_{i,j}^{n} + r_x \left(u_{i+1,j}^{n} - 2u_{i,j}^{n} + u_{i-1,j}^{n} \right) + r_y \left(u_{i,j+1}^{n} - 2u_{i,j}^{n} + u_{i,j-1}^{n} \right)$$

$$r_x = \frac{\alpha \, \Delta t}{\Delta x^2}, \qquad r_y = \frac{\alpha \, \Delta t}{\Delta y^2}$$

Each bracket represents the 1D directional difference.

Everything seems the same, but we need to study stability.

Stability of FTCS in 2D

Let's substitute a Fourier mode with one wavenumber per direction:

$$u_{i,j}^n = G^n e^{\mathrm{i}(kx_i + ly_j)}, \qquad \theta_x = k\,\Delta x, \qquad \theta_y = l\,\Delta y$$

$$\Longrightarrow \qquad G = 1 - 4r_x \sin^2\frac{\theta_x}{2} - 4r_y \sin^2\frac{\theta_y}{2}$$

G is real and its smallest value is at $\theta_x = \theta_y = \pi$, so $|G| \leq 1$ requires:

$$r_x + r_y \leq \tfrac{1}{2} \qquad \Longrightarrow \qquad \Delta t \leq \frac{\Delta x^2}{4\alpha} \quad \text{when } \Delta x = \Delta y$$

Remember that in 1D the limit was $\Delta x^2/2\alpha$.

Each dimension we add halves the time step again, so in d dimensions the limit is $\Delta x^2/2d\alpha$.

That cost is what motivates the implicit and split methods (see Tannehill 4.2.10,11).

The 2D Linear Advection Equation

In a 2D advection problem, we now carry a quantity with a constant velocity that has two components:

$$\frac{\partial u}{\partial t} + c_x \frac{\partial u}{\partial x} + c_y \frac{\partial u}{\partial y} = 0 \qquad \Longrightarrow \qquad u(x, y, t) = u_0(x - c_x t,\ y - c_y t)$$

The profile is translated along the vector (c_x, c_y) without change of shape.

Differencing backward in each direction, for $c_x > 0$ and $c_y > 0$:

$$u_{i,j}^{n+1} = u_{i,j}^{n} - C_x \left(u_{i,j}^{n} - u_{i-1,j}^{n}\right) - C_y \left(u_{i,j}^{n} - u_{i,j-1}^{n}\right)$$

We have 2 Courant numbers:

$$C_x = \frac{c_x \Delta t}{\Delta x}, \quad C_y = \frac{c_y \Delta t}{\Delta y}$$

The CFL Condition in 2D

Let's substitute the Fourier mode into the upwind scheme:

$$G = (1 - C_x - C_y) + C_x e^{-\mathrm{i}\theta_x} + C_y e^{-\mathrm{i}\theta_y}$$

This can be read as a weighted average.

If $C_x + Cy \leq 1$ the three coefficients are not negative and they sum to 1, so G is an average of 3 points of the unit circle and $|G| \leq 1$.

If the sum passes 1, $\theta_x = \theta_y = \pi$ gives $G = 1 - 2(C_x + Cy) < -1$:

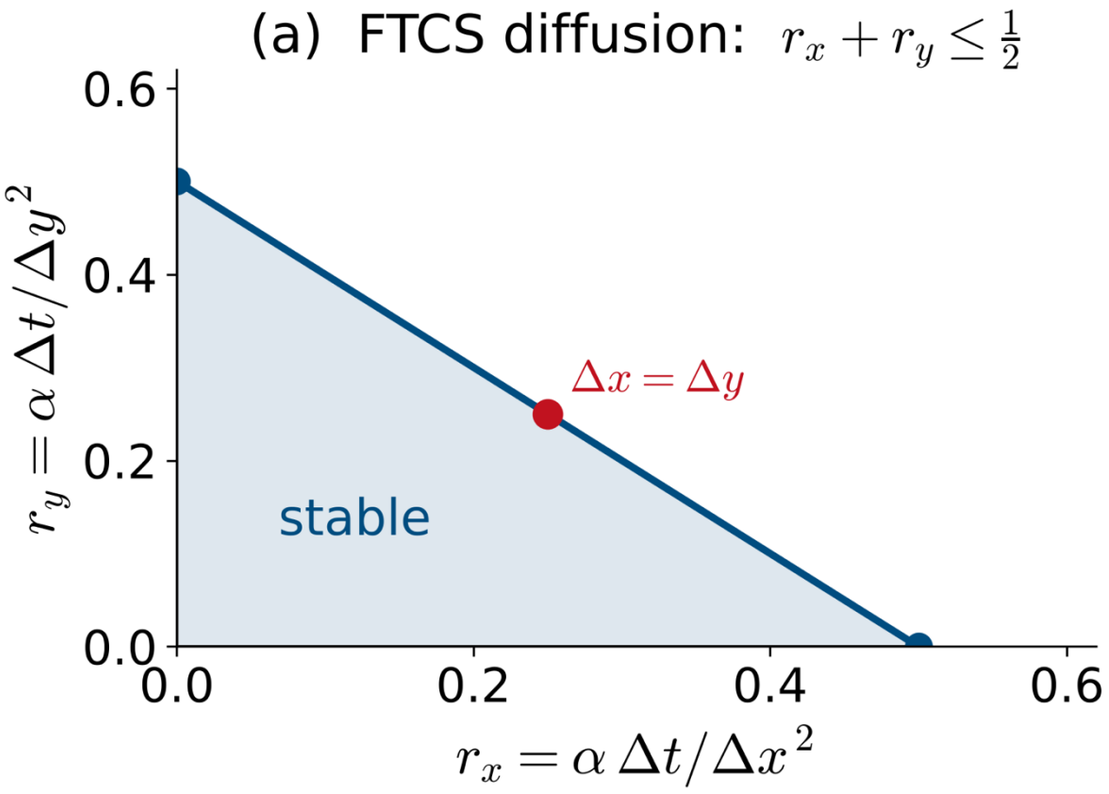
$$\boxed{C_x + C_y \leq 1}$$

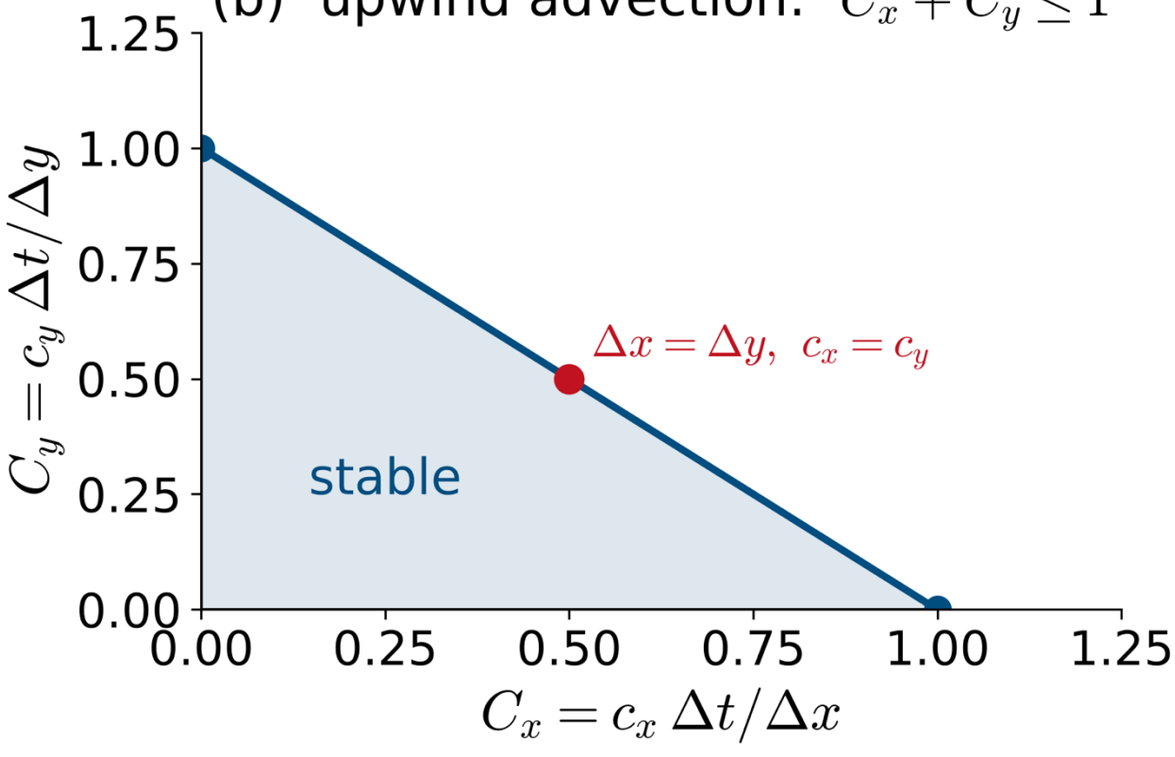
Notice that the Courant numbers add, they do not compete.

On a square grid with $c_x = c_y$ the time step is half of what the 1D condition would allow.

Stability Regions

Both conditions cut the same triangle out of the first quadrant:



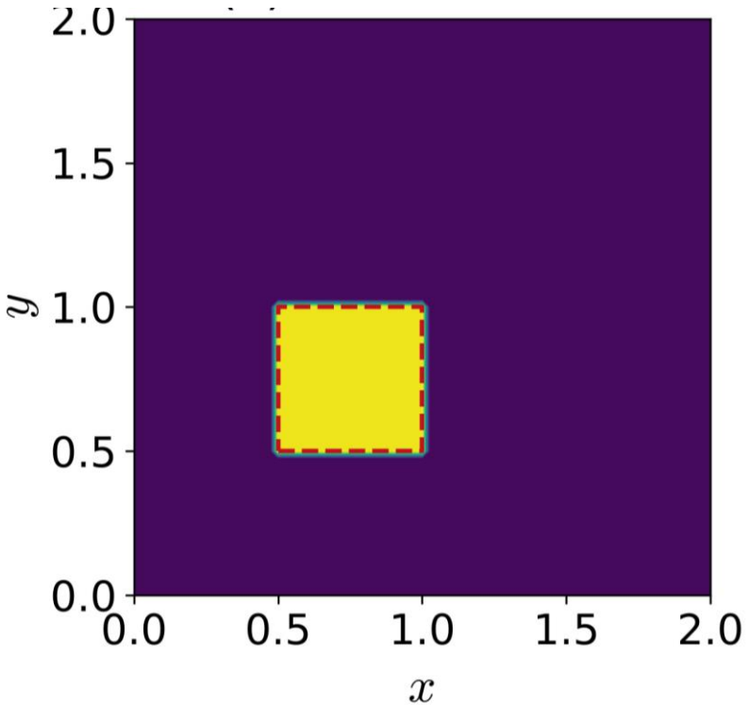


The red point marks a square grid with equal speeds.

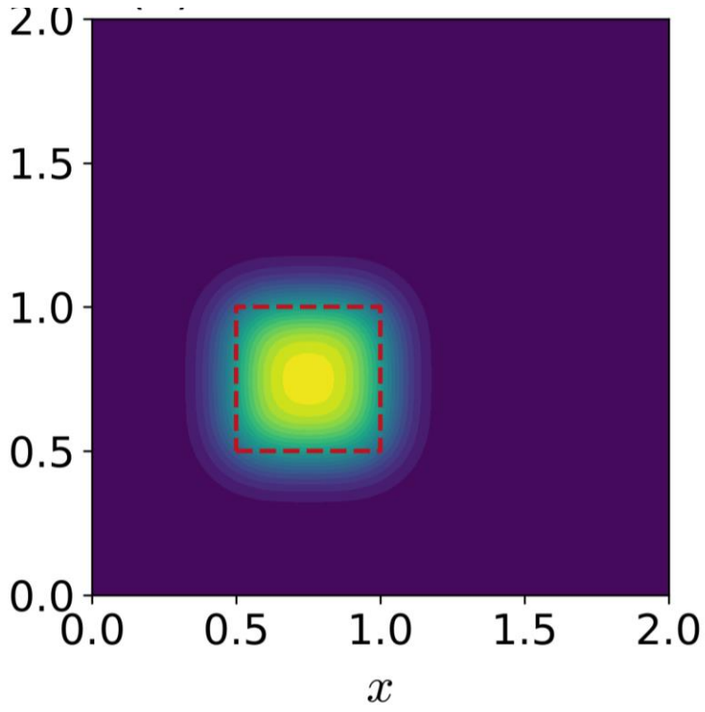
Solutions in 2D

The results for a square signal look like these:

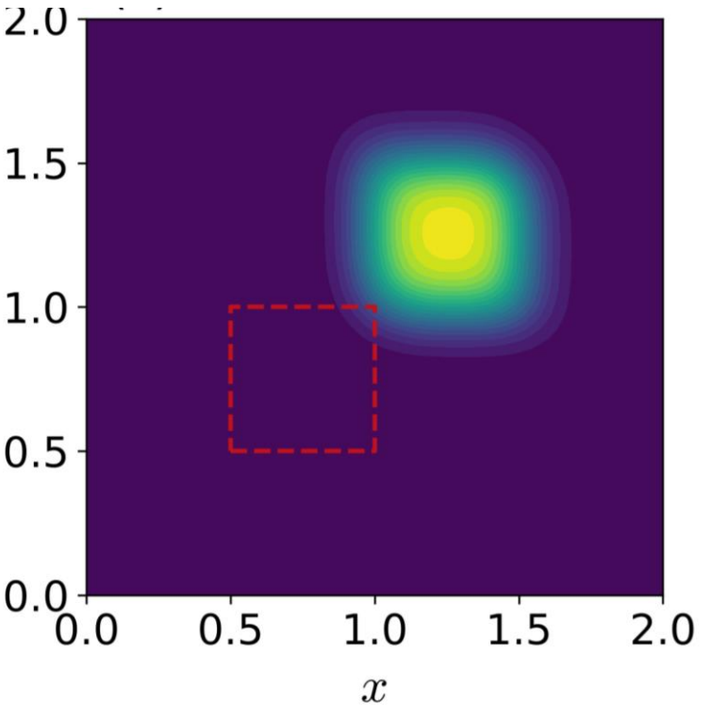
Initial top-hat function



Diffusion



Advection



Diffusion spreads the top-hat:

https://github.com/wbandabarragan/computational-fluid-dynamics/blob/main/notebooks/nb10_2d_diffusion.ipynb

Upwind advection moves it and smears it:

https://github.com/wbandabarragan/computational-fluid-dynamics/blob/main/notebooks/nb11_2d_advection.ipynb

The 2D Nonlinear Advection Equations

We replace the constant velocity by the solution itself.
In 2D the velocity has two components, so we need two coupled equations:

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} + v\,\frac{\partial u}{\partial y} = 0$$

$$\frac{\partial v}{\partial t} + u\,\frac{\partial v}{\partial x} + v\,\frac{\partial v}{\partial y} = 0$$

Both are marched together, each with the upwind scheme of the previous slide and with the local velocity in place of c_x and c_y.

Conservation Form: The divergence form now carries a flux vector:

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{F} = 0, \qquad \mathbf{F} = \left(\frac{u^2}{2},\ \frac{u^2}{2}\right) \qquad \Longleftrightarrow \qquad u_t + u\,u_x + u\,u_y = 0$$

For the non-conservative system the two forms are not the same equation:
$\partial_x(u^2/2) + \partial_y(uv) = u\,u_x + v\,u_y$ only when $\nabla \cdot V = 0$.
We therefore use the scalar flux above for the comparison that follows.

Jump Condition:

In 2D the Rankine-Hugoniot condition picks up the normal of the front:

$$s\ [u] = [\mathbf{F}] \cdot \mathbf{n}$$

Let's take a front whose normal points along (p, q), so that u depends on the single variable $\zeta = px + qy$.

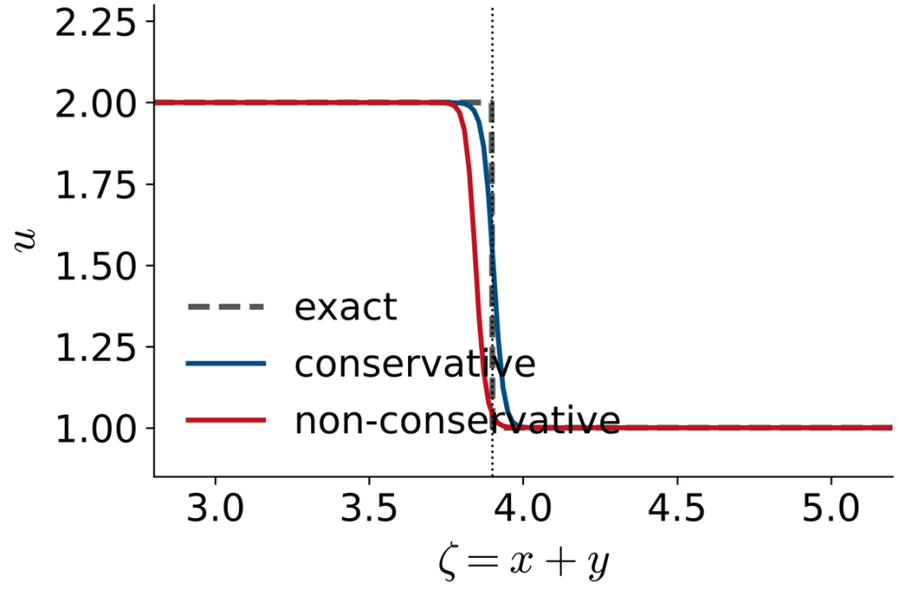
Putting $F = (u^2/2, u^2/2)$:

$$\boxed{s_\zeta = (p+q)\,\frac{u_1 + u_2}{2}}$$

The same jump travels at a different speed depending on how it is oriented.

Non-conservative approaches still place the shock at the wrong position and gives the wrong speed.

See plot on the right for the shock front along a diagonal.



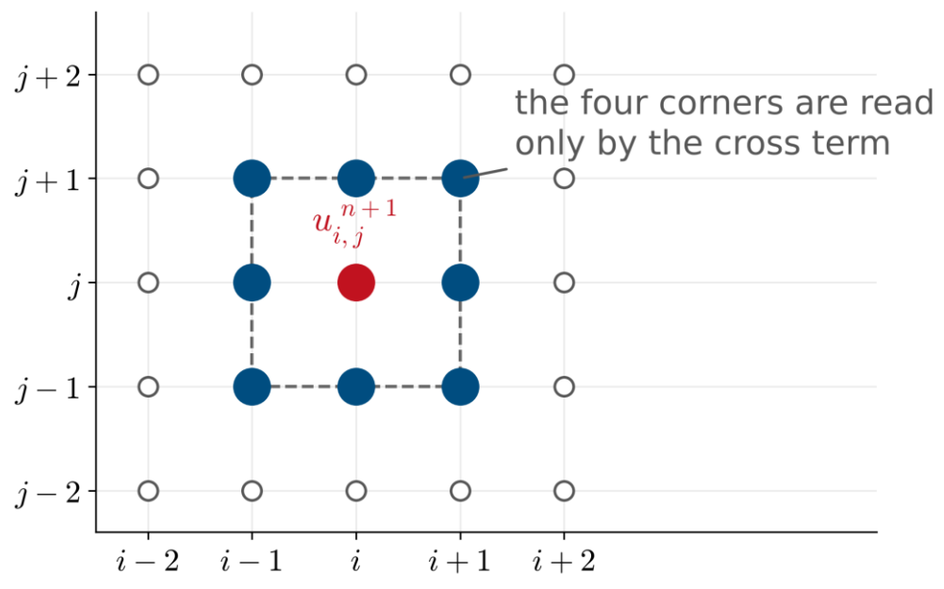
2D Lax-Wendroff Scheme

We expand in time to second order, taking both time derivatives from the equation:

$$u_{i,j}^{n+1} = u_{i,j}^{n} + \Delta t \frac{\partial u}{\partial t} + \frac{\Delta t^2}{2} \frac{\partial^2 u}{\partial t^2} + \mathcal{O}(\Delta t^3)$$

Centred differences in space then give the scheme:

$$\begin{aligned} u_{i,j}^{n+1} = u_{i,j}^{n} &- \frac{C_x}{2}\left(u_{i+1,j}^{n} - u_{i-1,j}^{n}\right) - \frac{C_y}{2}\left(u_{i,j+1}^{n} - u_{i,j-1}^{n}\right) \\ &+ \frac{C_x^2}{2}\left(u_{i+1,j}^{n} - 2u_{i,j}^{n} + u_{i-1,j}^{n}\right) + \frac{C_y^2}{2}\left(u_{i,j+1}^{n} - 2u_{i,j}^{n} + u_{i,j-1}^{n}\right) \\ &+ \frac{C_x C_y}{4}\left(u_{i+1,j+1}^{n} - u_{i+1,j-1}^{n} - u_{i-1,j+1}^{n} + u_{i-1,j-1}^{n}\right) \end{aligned}$$



Stability: the Fourier analysis gives $C_x + Cy \leq 1/\sqrt{2}$, tighter than the upwind limit of 1. Thus, on a square grid with equal speeds, the CFL number should be $C \leq 0.354$.

2D Lax-Wendroff: Cross Derivatives and Splitting

As we saw, there is a crossed term.

$$\frac{\partial^2 u}{\partial t^2} = \left(c_x \frac{\partial}{\partial x} + c_y \frac{\partial}{\partial y} \right)^2 u = c_x^2 u_{xx} + 2 c_x c_y \, u_{xy} + c_y^2 u_{yy}$$

The cross term has no one-dimensional ancestor and needs a nine-point stencil.

One way around it is to split the step, marching in x and then in y with one-dimensional operators:

$$u^* = L_x(\Delta t)\, u^n, \qquad u^{n+1} = L_y(\Delta t)\, u^*$$

Tannehill 4.2.11 and 4.5.8 develop this as the fractional-step and time-split methods.

Summary

- We apply the 1D formulae in each direction. The diffusion stencil gains 2 points.

- The FTCS scheme for the heat equation in 2D is stable for $r_x + r_y \leq 1/2$, and the limit tightens to $\Delta x^2/2d\alpha$ in d dimensions.

- The upwind advection scheme is stable for $C_x + Cy \leq 1$. The Courant numbers add.

- The nonlinear problem becomes a coupled pair, one equation per velocity component.

- The jump condition carries the normal of the front, $s[u] = [F] \cdot n$, so the speed depends on the orientation.

- The conservative form reaches that speed at every orientation; the non-conservative form is slow at all of them.

- Second-order schemes bring a cross derivative, which is why the step is often split into 1D sweeps.